(1.) Course Title: INFPHD447 Formal and Semi-Formal Approaches for Information Security	**Credits: 6**
Course **Classification: Optional**	
Theoretical or Practical Nature of the Course, *'training character'*[13]: theory-practice ratio 100% (credit%)	
Type of Class: lecture and **number of hours**: **2/week** in the given semester, *(If the course is not (only) taught in Hungarian, then **the language**: **English**)* **Additional** (specific**) methods and characteristics**[2] used in delivering the knowledge *(if any)*: Continuous mid-year assessment, small lectures based on literature review.	
Assessment Method (Exam / Midterm Grade / **Other**[3]): colloquium **Additional** (specific) **methods**[4] applied in knowledge assessment *(if any)*: LMS	
Course Placement in Curriculum (Semester Number): -	
Prerequisites (*if any*): -	

Course Description: **A concise yet informative description of the knowledge** to be acquired.

FORMAL METHODS OF INFORMATION SECURITY
1.1 Formal models of information security
1.2 Formal models of information security requirements needs.
1.3 A Trace-Based Model.
1.4 Verification procedures
Concepts and notations,
2.1 System Model
2.1.1 Monitoring, audit trail
2.1.2 Event-based systems
2.1.3 State-event systems
2.2 Specification of system characteristics
2.3 Trace characteristics, closure properties
2.4 Methodology for specifying secure information flow.
2.5 Other specification formalisms.
2.6 Model driven information security
2.7 Pattern-based approaches
3. INFORMATION SECURITY ARCHITECTURE
3.1. PUBLIC KEY INFRASTRUCTURE (PKI) SERVICES AND THEIR CHARACTERISTICS
3.2. CERTIFICATE-BASED PERSONAL IDENTIFICATION
3.3. TIME STAMP SERVICE
3.4. SOLUTIONS REQUIRED BY SECURE LINE COMMUNICATIONS
3.5. COMMUNICATION AND NETWORK SECURITY ASPECTS
3.6. GENERAL REQUIREMENTS FOR SOFTWARE ARCHITECTURE SERVICES
3.7. SECURITY ARCHITECTURE REQUIREMENTS
3.8 SECURITY REQUIREMENTS FOR SOA, WEB SERVICES
3.8.1. WS-Security
3.8.2 WS-Trust
3.8.3. XACML
3.8.4. Drawing logical conclusions about security policies
3.9. BASICS OF A STANDARD INFORMATION ARCHITECTURE FOR SYSTEMS

3.10. SINGLE SIGN-ON AND FEDERATED IDENTITY PROTOCOL
3.10.1 Basic operating model
3.10.2. Advantages
3.10.3 SSO over the Internet and federated identification
Models 3.10.4: Use of Internet services with single sign-on
3.11. SAML-BASED SSO USE CASES
3.11.1. Participants in SAML dialogues
3.11.2 Use case of single sign-on via Internet/Web/Web
3.11.3 Use case for federated identification
3.11.4. Federation
3.12. ARCHITECTURAL SOLUTIONS FOR REPRESENTING ENTITLEMENT RELATIONSHIPS
3.12.1 Role and Entitlement Design
3.12.2 Workflow model in scenario concept
3.12.3 Role design, organisation
4. Blockchain technology and applications in Information Systems
4.1 Blockchain patterns.
4.2 IT and cybersecurity requirements for auditing and certifying blockchain systems.
5. The role of artificial intelligence and machine learning in IT and cybersecurity.

2-5 Most Important Required and Recommended Literature

List of required and recommended literature (notes, textbooks) with bibliographic data (author, title, publication details, (possibly pages), ISBN).

1. Stamp, M. and Jureček, M. eds. 2025. *Machine Learning, Deep Learning and AI for Cybersecurity*. Springer Nature Switzerland.
2. Beckers, K. (2015). *Pattern and Security Requirements: Engineering-Based Establishment of Security Standards*. Cham: Springer International Publishing. doi: 10.1007/978-3-319-16664-3
3. Bidgoli, H. (2006). *Handbook of information security*. Hoboken, NJ: Wiley.
4. Hafner, M., & Breu, R. (2009). *Security engineering for service-oriented architectures* (pp. I-XVI). Berlin, Heidelberg: Springer.
5. Site, E., Cuts, R., & In, S. (2008). Security Engineering: A Guide to Building Dependable Distributed Systems. *2ed Editio*, 239-274.
6. Gertz, M., & Jajodia, S. (2007). *Handbook of database security: applications and trends*. Springer.
7. Macartney, L. A. (2005). *Information Security Harmonisation: Classification of Global Guidance*. Information Systems Audit and Control Association.
8. Killmeyer, J. (2002). *Information security architecture: an integrated approach to security in the organization*. CRC Press.
9. D. Garlan, M. Shaw, An Introduction to Software Architecture, Advances in Software Engineering and Knowledge Engineering, Volume I, World Scientific, 1993.

List of Prescribed Professional Competencies and Competency Elements

(knowledge, skills, etc., KKK Section 8) that the course significantly contributes to developing (highlight the most important ones).

a) Knowledge

- Has a research-level understanding of the general and specific characteristics, major directions, and precisely developed boundaries of the subject area within the discipline, as well as the agreed and debated interrelations.
- Creatively comprehends the interrelations, theories, and conceptual frameworks and terminologies of the given field of study.1
- Possesses research methodology knowledge necessary for independent research in the given discipline/field.

b) Skills

- Capable of creative analysis of the field, synthesizing comprehensive and specific interrelations with a new perspective, and conducting appropriate evaluative and critical activities.
- Able to apply and further develop the specialized learning and problem-solving methods of the field.
- Can creatively elaborate on the novel, previously unknown practical implications of theoretical issues.
- Capable of planning and executing new projects, conducting research in the field, and developing new techniques and approaches.
- Able to identify unforeseen professional problems and uncover the detailed theoretical and practical background necessary for their resolution at a research level.
- Can construct and communicate new relations relevant to the field, as well as comprehensive interrelations significant for personal and communal existence.

c) Attitudes

- Represents and further develops the relations that contribute to the process of human self-creation based on the field's characteristics.
- Possesses an interest and learning ability that enable identifying and solving currently unpredictable research problems in the field.
- Demonstrates a firm professional commitment, a continuous dedication to exploring new approaches, and an acceptance of the necessity for persistent effort.

d) Autonomy and Responsibility

- Builds and initiates new knowledge areas with creative independence and proposes new practical solutions.
- Participates in formulating theoretical and practical questions with a leadership role and high-level cooperation.
- Engages as an equal discussion partner with professionals in the field.

[1] **Section 108**, Article 37 of the National Curriculum Act (Nftv.) 37. lesson: a session (lecture, seminar, exercise, consultation) requiring the personal involvement of the instructor to fulfil the academic requirements specified in the curriculum, the duration of which is at least forty-five minutes and not more than sixty minutes.

[2] e.g. case studies, role play, thematic presentations, etc.

[3] e.g. continuous debriefing, mid-year report

[4] e.g. request for case studies, thematic papers, essays, business, organisational plans, etc.

• Takes responsibility for raising and addressing new ethical questions concerning theoretical and practical aspects of the profession.
Course responsible (name, position, academic degree): **Dr. Bálint Molnár, habil. Associate Professor, PhD**
Instructor(s) involved in teaching the course (if any) (name, position, academic degree): **Dr. Khawla Bouafia, Assistant Professor**